International Journal of Distributed Systems and Technologies
Volume 16 • Issue 1 • January-December 2025

Assessing Forest Quality Using Multi-Source Satellite Remote Sensing Data:

A Case Study in Western Beijing's Mountainous Regions

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ABSTRACT

This study uses Sentinel satellite data to estimate forest quality over a large area, focusing on Beijing. By combining ground survey data with remote sensing, a random forest model predicts forest parameters. The results show a correlation coefficient of 0.60-0.76 and a relative root mean square error of 0.09-0.39. Average tree height and diameter at breast height (DBH) had the highest accuracy (75%-80%), followed by canopy density and plant number density (68%-75%). The spatial agreement between predicted and actual forest quality indicates the model's effectiveness.

KEYWORDS

Sentinel Data, Beijing, Forest Quality, Stand Factor, Random Forest

INTRODUCTION

In the context of global climate change, forests, as critical terrestrial ecosystems, play an essential role in mitigating carbon dioxide emissions and attaining carbon neutrality (Ghasemi et al., 2024; Xu, 2023). The concept of forest quality encompasses a multifaceted assessment that integrates ecological, societal, and economic dimensions. Research into forest and site quality enhances our understanding of forest functionality and valuation, facilitates the formulation of judicious management practices for foresters, and fosters the modernization and sustainability of the forestry sector (Lehmler et al., 2023). Traditional methods for assessing forest health are predominantly based on on-site investigations, which establish various indicators to evaluate forest attributes. While these approaches offer high accuracy, they are labor-intensive and resource-demanding, thus limiting their applicability over extensive areas. To address this limitation, remote sensing technologies provide a viable alternative for large-scale and rapid forest health assessments.

DOI: 10.4018/IJDST.383047

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This study focused on stand-level attributes to assess the ecological functions of forests. By leveraging state-of-the-art remote sensing data and integrating it with ground-based survey information, a regression model was constructed to predict stand factors within the study area. A novel aspect of this approach was the application of the random forest (RF) algorithm to quantify the relative significance of individual feature variables. By analyzing the sensitivity of these variables to forest and site quality, stand quality and its components were evaluated, clarifying their spatial distribution patterns and interrelationships.

The main contribution of this research lies in having achieved high-precision estimates of forest quality using Sentinel-1 and Sentinel-2 satellite data through an optimized RF regression model. It also acknowledged the assumptions and limitations inherent in the study, such as the potential impact of relying on data from specific regions on the model's generalizability and the fact that only static analysis was conducted, without fully capturing the dynamic processes of forest and site quality.

The article begins by introducing the motivation and background of the study, followed by an overview of the proposed methodological framework. Detailed descriptions of variable selection and model construction processes are provided, including the use of the Boruta algorithm for feature selection and the RF algorithm for building regression models. Additionally, the paper elaborates on data acquisition and preprocessing, ground sample data collection, and feature factor extraction. Results demonstrated the inversion accuracy of different forest parameters and validated the effectiveness of the proposed method by comparing it with other regression models. Conclusions summarize the findings and propose future directions, emphasizing the importance of integrating multi-source remote sensing imagery, unmanned aerial vehicles (UAVs), and light detection and ranging (LiDAR) technology to improve the precision of forest quality assessments.

RELATED RESEARCH

De Moura Fernandes et al. (2023) evaluated the mapping performance of spectral and textural variables derived from MSI/Sentinel-2 imagery for vegetation segments in the Caatinga biome, northeastern Brazil. The study determined multiple indicators, including raw reflectance values, which yielded a high estimation accuracy. Bera et al. (2023) analyzed the efficacy of Sentinel-2 data in predicting canopy cover, revealing that its predictive power was inferior to that of a full Landsat-8 model, as Sentinel-2 distributed variable importance across a broader set of features. Muñoz et al. (2022) utilized a normalized difference index based on Landsat imagery to produce original forest cover maps, subsequently incorporating improved Sentinel-1 and ALOS PALSAR data fusion. This approach achieved a classification accuracy of 0.946 for forested areas and 0.92 for non-forest regions.

To improve the precision of forest parameter inversion, Zhen (2022) applied super-resolution reconstruction of Sentinel-2 images, combined with support vector machine (SVM)-recursive feature elimination, to select optimal variables for estimating canopy nitrogen content. The regression model developed for mangrove canopy nitrogen content prediction exhibited a fitting accuracy comparable to that of a resampled model. Lasko et al. (2024) employed a RF classifier, utilizing Sentinel-2 and low-density time-series images for automatic data training and the generation of a near-global land cover classification framework. The evaluation indicated that the regional adaptation threshold for land cover could accurately delineate land cover types under 9-level, 6-level, and 5-level classification schemes.

Zou et al. (2023) utilized nonlinear least squares and Bayesian methods to analyze dominant height data from *Pinus massoniana* stands, developing an approach that demonstrated superior fit for stand quality and potential productivity predictions. Wambsganss et al. (2022) assessed the influence of tree species mixing on fine root litter quality degradation, considering the interplay between species composition and site conditions. Their findings suggested that while species mixtures can modify fine root litter quality loss across large environmental gradients, their impact was relatively minor compared to that of environmental factors. Dong et al. (2022) constructed a site quality model using the

algebraic difference method, employing a rough set model to identify key site factors affecting the quality grade of Chinese fir plantations. A site quality classification model was then developed using RF, achieving training and testing accuracies of 92.47% and 78.46%, respectively. Vaca et al. (2022) used linear models and analysis of covariance to create a site quality map, indicating that 86% of production forests within the study area were categorized as having low productivity, with the actual growth height of peach blossom heartwood being lower than predicted.

Roshani et al. (2023) leveraged remote sensing to assess forest health, integrating various layers via a weighted overlay method to produce a comprehensive forest health map. The results highlighted that the largest area was occupied by moderately healthy forests, followed by those in good health (Rawat & Saini, 2023).

In summary, traditional forest health assessments have been predominantly based on field investigation data, which are labor-intensive, time-consuming, and limited in spatial coverage. With the increasing utilization of remote sensing technologies, these limitations have been mitigated through the advantages of large-scale, rapid, and cost-effective monitoring. Consequently, this study explored the application of Sentinel-based remote sensing for the estimation of forest quality. By applying the RF feature importance analysis, the aim was to quantitatively characterize the relative significance of various parameters in Sentinel data, thus enabling high-precision remote sensing estimates of forest quality.

This study was primarily conducted in the mountainous regions of western Beijing, encompassing an area defined by geographic coordinates N39° 37' to 40° 39' and E115° 55' to 117° 08'. The total mountainous area within Beijing spans approximately 10,400 square kilometers, constituting 62% of the city's overall territory. This region represents a significant portion of rocky mountainous terrain in North China and holds considerable national importance (Geng et al., 2018; Ma et al., 2023). Geographically, it is in the northern part of North China, bordering the Taihang Mountains to the west, the Shanxi Plateau to the north, and the Inner Mongolian Plateau to the northwest. Due to its strategic location, the area serves as a vital ecological buffer for the socio-economic development of North China.

In the mountainous areas of Beijing, broad-leaved shrubs are predominant, while broad-leaved and coniferous forests are also prevalent. Historically, the forest coverage in Beijing prior to 1949 was merely 2.59%. However, through sustained reforestation efforts, extensive plantations of oak, cypress, pine, poplar, locust, and larch have been established. According to the distribution map of forest resources in Beijing (2014–2018), the total forested area amounts to 10,710 km^2, with a forest coverage rate of 43.77%. Among the dominant tree species, oak (*Quercus spp.),* lateral cypress (*Platycladus orientalis*), and Chinese pine (*Pinus tabuliformis*) collectively occupy 26.2% of the total forested area. These artificial forests, largely due to anthropogenic influences, are characterized by a predominance of medium and small-sized, monospecific stands, which exhibit limited species diversity and stability, thereby posing potential ecological risks.

The Capital Region Ecological Station falls within the warm-temperate deciduous broad-leaved and mixed conifer-broadleaved forest zone, with a forest coverage of 96.4%. This area is predominantly composed of plantations established post-1950, with sawtooth oak (*Quercus acutissima*) being the primary afforestation species. Given this context, the current research focused on two representative dominant tree species: sawtooth oak (*Quercus acutissima*) and another unspecified oak species (presumably also *Quercus acutissima* or a different oak species, as the text mentions it twice), as the subjects of the study (Ota, 2012; Zhou et al., 2021).

METHODOLOGY

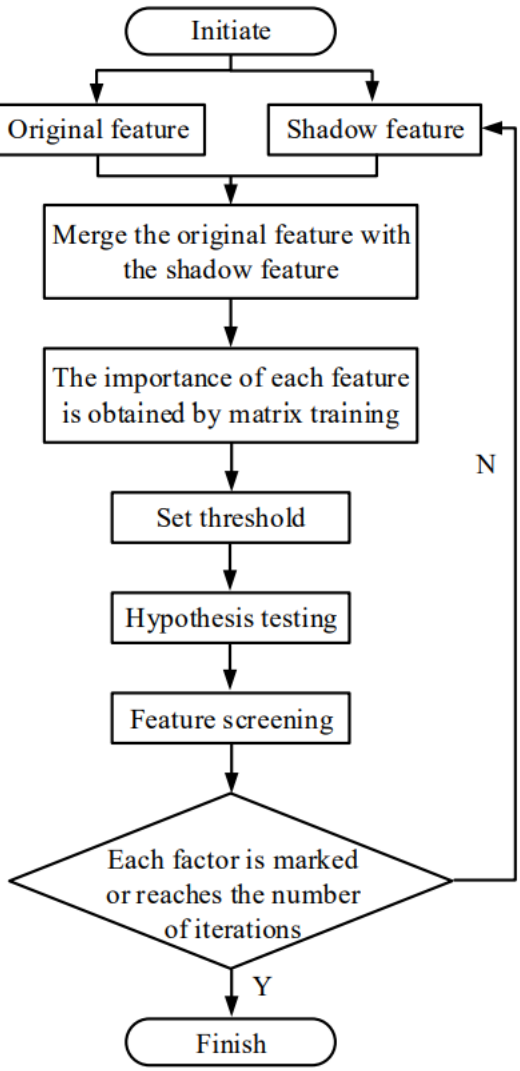
Variable Selection and Model Construction

In the construction of a forest quality assessment model, variable selection is a critical step. An excessive number of variables can lead to issues such as prolonged computational processing and

diminished model accuracy. It was imperative, therefore, to identify the optimal subset of variables during the modeling process. Selecting appropriate variables not only enhanced the utilization of Sentinel data but also improved computational precision while reducing complexity.

One effective method for feature selection is the Boruta algorithm, which facilitates dimensionality reduction and simplifies model complexity by identifying the most relevant and informative features (Duan et al., 2023). This optimization in turn enhances predictive performance and the capacity to handle large-scale datasets. Consequently, the Boruta algorithm was employed to select features for various forest parameters. Different indicators were chosen as independent variables for the inversion of forest factors. The application of the Boruta algorithm followed a structured process, as illustrated in Figure 1.

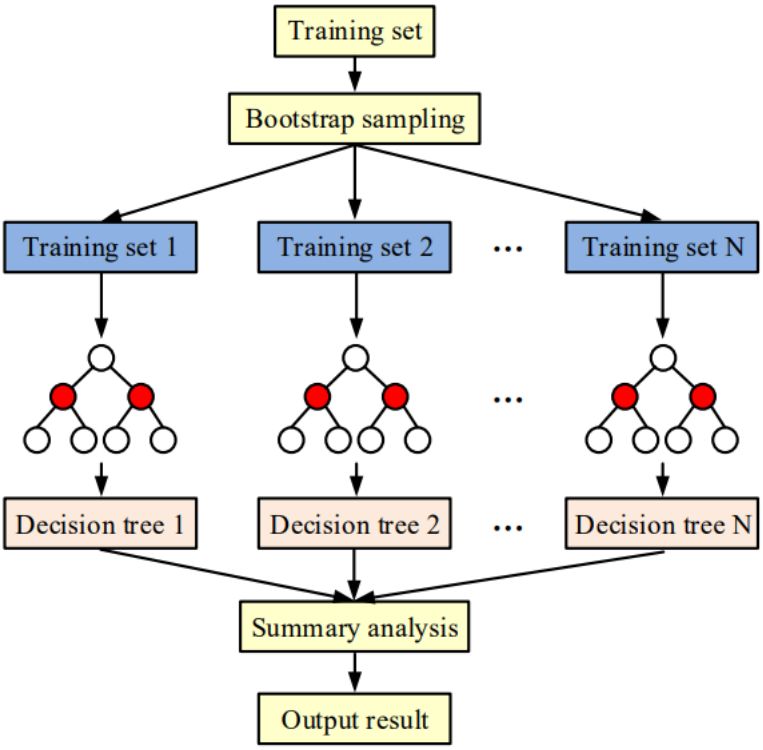
Figure 1. Boruta algorithm application flow



Initially, shadow features were generated. For each original feature, the values were randomly permuted to create a corresponding shadow feature, which served as a reference for assessing the importance of the actual features. These shadow features were then concatenated with the original features to form an extended feature matrix. The Boruta algorithm subsequently input this new feature matrix and computed the importance scores for both the original and shadow features. The importance of each feature was compared against its shadow counterpart. If the importance score of an original feature significantly exceeded that of its shadow feature, the feature was deemed important and retained. Conversely, if the importance score was notably lower than that of the shadow feature, the feature was marked as unimportant and excluded from further consideration. Through iterative rounds of comparison, the algorithm effectively identified and retained the truly significant features, ultimately outputting the refined set of selected features.

The RF algorithm has the advantages of a simple structure, easy implementation, and resistance to data loss and imbalance, making it particularly suitable for problems with small and large sample features (Amankulova et al., 2023; Chen et al., 2024; Dallahi et al., 2023; Pandey et al., 2021). Therefore, the study selected the RF algorithm to construct a regression model, as shown in Figure 2. In the process of classification and regression, the sample set was randomly selected, and the bootstrap method used to randomly select training subsets, with each sample set not exceeding two thirds of all samples. Based on the extracted training subsets, classification and regression trees were established. Then, a forest consisting of these classification and regression trees was generated. For each node of each classification and regression tree, a subset of variables was randomly selected from all input variables in the training set for prediction. This method used the prediction results of the regression tree to vote on new samples and determine their size. The classification and regression model of RF was implemented using RStudio software.

Figure 2. RF algorithm example



Data Acquisition and Preprocessing

Regarding satellite image data features, the satellite image data included Sentinel-1 synthetic aperture radar (SAR) images and Sentinel-2 multispectral imagery. Key features extracted from these datasets included data from Sentinel-1 SAR. Texture information was derived using the gray level co-occurrence matrix method, with eight texture features calculated in four distinct directions under VV and VH polarizations: contrast, mean, correlation, variance, dissimilarity, homogeneity, entropy, and angular second moment.

On June 25, 2023, a C-band dual-polarization SAR image was acquired from the Sentinel-1 satellite with a processing level of ground range detected. This dataset underwent several preprocessing steps to ensure its accuracy and usability for subsequent analysis. Initially, orbit files were applied to correct the satellite's position and velocity, ensuring precise geolocation information. Following this, edge noise removal was conducted to eliminate artifacts present at the boundaries of the images, enhancing overall image quality.

Further refinement involved thermal noise reduction, aimed at mitigating the effects of sensor-generated noise inherent in SAR imagery. To address the granular noise characteristic of SAR images, speckle filtering was employed, improving the clarity and interpretability of the data. Radiometric calibration converted digital numbers into backscatter coefficients, providing standardized measurements essential for comparative analysis across different scenes or over time.

Terrain correction adjusted the images to account for topographic variations, ensuring that the radar backscatter accurately reflected surface properties rather than terrain-induced distortions. Subsequently, geocoding aligned the image with a geographic coordinate system, facilitating spatial

referencing and integration with other geospatial datasets. Lastly, image cropping focused on the area of interest, allowing for efficient processing and analysis by eliminating unnecessary data.

All preprocessing tasks except for image cropping, which is performed using ENVI 5.3 software, were executed within the sentinel application platform. This comprehensive preprocessing protocol ensured that the final dataset was optimized for evaluating forest quality and stand characteristics, integrating seamlessly with additional data sources such as ground sample data and multispectral imagery from Sentinel-2.

Sentinel-2 Multispectral Data

On August 10, 2023, Level-2A products from the Sentinel-2 satellite were obtained, providing data at three ground resolutions: 10 m, 20 m, and 60 m. For detailed analysis, ten spectral bands (B) were selected based on correlation analysis. These included B2 (blue) with a wavelength of 490 nm and a resolution of 10 m, B3 (green) at 560 nm also with 10 m resolution, and B4 (red) at 665 nm again with 10 m resolution. Additionally, B5 (vegetation red edge) at 705 nm, B6 (vegetation red edge) at 740 nm, B7 (vegetation red edge) at 783 nm, and B8A (vegetation red edge) at 865 nm all had a resolution of 20 m. B8 (near infrared) operated at 842 nm with a resolution of 10 m, while B11 (shortwave infrared) at 1610 nm and B12 (shortwave infrared) at 2190 nm shared a resolution of 20 m.

Table 1 presents the band information from the Sentinel-2 multispectral imager utilized for processing and analysis.

Table 1. Sentinel-2 multi-spectral imager band information

Band (B) abbreviation	Band	Central wavelength (*nm*)	Spatial resolution (*m*)
B1	Coastal aerosol	443	60
B2	VIUE	490	10
B3	Green	560	10
B4	Red	665	100
B5	Red edge of vegetation	705	20
B6		740	20
B7		783	20
B8	Near infrared	842	10
B8A	Vegetation red edge	865	20
B9	Steam	945	60
B10	Shortwave infrared cirrus clouds	1375	60
B11	Short-wave infrared	161	20
B12		219	20

To support forest biomass inversion, eight primary vegetation indices were extracted from these spectral bands. These indices include the enhanced vegetation index, simple ratio, soil adjusted vegetation index, normalized difference vegetation index, atmospherically resistant vegetation index, modified soil adjusted vegetation index, difference vegetation index, and non-linear vegetation index. Each of these indices served to highlight different aspects of vegetation health and density, facilitating a comprehensive assessment of forest conditions.

Preprocessing steps for these datasets ensured that the data was optimized for analysis. Initially, each band underwent resampling to achieve a uniform 10-meter resolution, enhancing spatial consistency across all bands. Radiometric and atmospheric corrections were applied to account for variations in illumination and atmospheric conditions, ensuring that the measurements accurately reflected surface properties. Following these corrections, the atmospherically corrected images were converted into formats compatible with ENVI 5.3 for further analysis. Finally, mosaicking, cropping, and masking processes isolated the relevant data specific to the study area, streamlining the dataset for efficient processing and analysis. This meticulous preprocessing protocol ensured that the final dataset was robust and reliable, suitable for integration with other geospatial data sources such as SAR imagery from Sentinel-1.

Combined Feature Set

The final dataset integrated both ground sample data and satellite image data features to provide a comprehensive assessment of forest quality.

Spatial Alignment

This ensured that all datasets were aligned by using the same geographic coordinate system (e.g., WGS 84). This involved cropping satellite images to match the study area and extracting corresponding values for each ground sample point.

Feature Extraction

This process extracted relevant features from satellite imagery, including soil adjusted vegetation index, texture variables (such as correlation), and specific multispectral bands (such as B2, B3, B4, B8).

Integration

This combined extracted satellite features with ground sample data attributes (such as canopy density, mean DBH, standing volume) into a single dataset. Each record in this dataset had both types of features, allowing for a more holistic analysis.

Modeling

This study used this combined dataset to train machine learning models like RF regression. For instance, the Boruta method was employed to select significant features contributing to forest stand factors. Optimal parameters for RF models were determined based on minimizing RMSE.

For evaluating the accuracy of the RF regression model, a random sampling approach was adopted, dividing the dataset into 70% for training and 30% for validation. The effectiveness of the forest factor inversion and the selection of the optimal inversion model were assessed using metrics such as the coefficient of determination (R^2), RMSE, and relative root mean square error (rRMSE). Data analysis was conducted using *R* language, with all evaluation indices calculated according to Equation (1).

$$\begin{cases} R^2 = 1 - \dfrac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \\ RMES = \sqrt{\dfrac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{n}} \\ rRMES = \dfrac{\sqrt{\frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{n}}}{\bar{y}} \times 100\% \end{cases} \tag{1}$$

In Equation (1), y_i signified the measured forest stand factor. $\hat{y}_i$ signified the predicted forest stand factor; $\bar{y}_i$ was the mean of the measured forest stand factor; n was the sample size.

Ground Sample Data

The ground sample data consisted of 2,681 field survey records collected between June and September 2022. These records included a comprehensive set of parameters that were essential for evaluating forest quality and stand quality. The key features from this dataset were:

- Geographical coordinates: precise location of each sample plot.
- Date of survey: time when the survey was conducted.
- Nomenclature of tree species: identification of tree species present in the plot.
- Mean tree height (m): average height of trees within the plot.
- Timber volume (m^3/ha): total volume of timber per hectare.
- DBH (cm):mean diameter of trees measured at BH (typically 1.3 meters above the ground).
- Canopy density: proportion of the forest floor covered by tree canopies.
- Predominant tree species: dominant species in the plot.
- Mean age of trees (years): average age of trees in the plot.
- Forest typology: classification of forest types based on structure and composition.
- Prevailing ecological conditions: environmental factors affecting the forest, such as soil type and moisture levels.

To ensure the integrity, uniformity, and stochastic representation of the estimated data, a stratified random sampling methodology was adopted. This approach facilitated the division of the forested area into five distinct strata based on an equal interval classification of the quality of the forest land as assessed *in situ*. Subsequently, a 10% proportional representation from each stratum was selected. Consequently, 268 sub-plots were chosen from the aggregate of 2,681 plots, forming an independent random sample. The characteristics of the sampled data are summarized in Table 2.

Table 2. Extract sample data information

Forest parameter	Radius	Mean value	Standard deviation	Coefficient of variation
Average DBH (cm)	7–25	13.04	2.87	0.24
Canopy density	0.2–0.9	0.56	0.12	0.26
Storage capacity (m^3/hectare)	9.63–154.72	63.52	31.77	0.47
Stand age	5–35	12.96	6.43	0.31
Average tree height (m)	3.5–15	7.68	3.02	0.25
Plant density (plant /hectare)	121–2056	1045.36	375.41	0.42

FOREST QUALITY ASSESSMENT

In the evaluation of forest health, the selection of appropriate indicators is paramount. This study adhered to the principles of scientific rigor, comprehensiveness, and practical applicability. It was grounded in the assessment of various forest attributes, from which a set of indicators for forest health evaluation had been derived. The factor analysis module within statistical package for the social sciences version (SPSS) 18.0 was utilized to investigate the inter-correlations among these indicators. Through an iterative process involving multiple analyses, the initial set of indicators was refined, resulting in a final selection of eight key metrics: average DBH, timber volume, mean tree height, canopy density, plant number density, aspect of slope, gradient of slope, and slope position. These indicators were subsequently quantified to facilitate further statistical analysis.

RESULTS AND ANALYSIS

Analysis of Forest Quality Evaluation System Results

The inter-correlations among various indicators, as determined through factor analysis, were examined to construct a forest quality evaluation index system specific to the Beijing area. The relative weights of each indicator within this system were calculated, and the comprehensive evaluation framework is presented in Table 3. According to the table, the mean DBH, mean tree height, canopy density, and plant density of the forest stand were assigned relative weights of 0.172, 0.199, 0.247, and 0.229, respectively. These four indicators collectively accounted for the highest proportion among the eight evaluated metrics. In contrast, the aspect, slope gradient, and slope position were attributed relatively lower weights, indicating their lesser influence on the overall forest quality assessment.

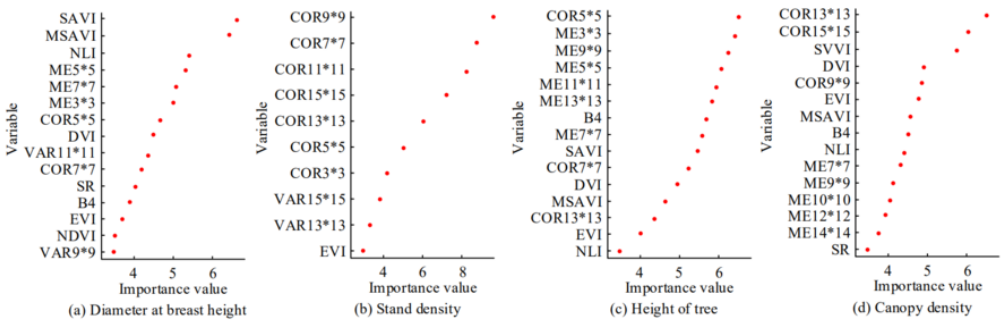
Table 3. Forest quality evaluation

Index group	Relative weight	Index	Relative weight	Global weight
Stand factor	0.759	Mean diameter breast height	0.172	0.172
		Standing volume	0.153	0.146
		Mean tree height	0.199	0.170
		Canopy density	0.247	0.143
		Plant density	0.229	0.137
Topographic factor	0.241	Aspect of slope	0.321	0.071
		Slope gradient	0.347	0.075
		Slope position	0.332	0.073

Analysis of the Relationship Between Forest Quality and Stand Factors

In the analysis of Sentinel data, the Boruta method was employed to select characteristic factors associated with forest stand attributes. The relative weights of each influencing factor were statistically analyzed to determine their significance. Forest stand factors played a crucial role in the selection and prioritization of various indicators. The explanatory variables and their importance for each forest stand factor are illustrated in Figure 3. Among the forest factors, the soil adjusted vegetation index exhibited the highest correlation coefficient with the mean DBH. Conversely, the texture variable correlation demonstrated the highest correlation coefficient with canopy density, forest density, and tree height.

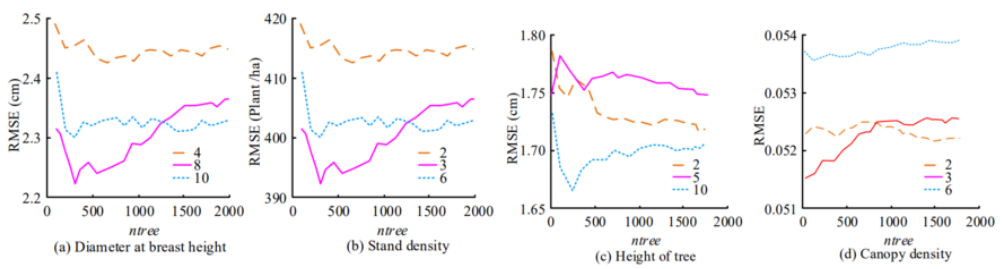
Figure 3. Analysis of the importance value of characteristic factors



The study revealed that, among the spectral parameters related to forest stand factors, texture variables were predominant. While these texture variables showed correlations with various forest stand elements, their association with other forest stand elements was relatively low, exerting a significant impact primarily on forest age and DBH. Overall, the analysis indicated a high correlation between texture parameters and forest stand factors, with a correlation coefficient exceeding 60%.

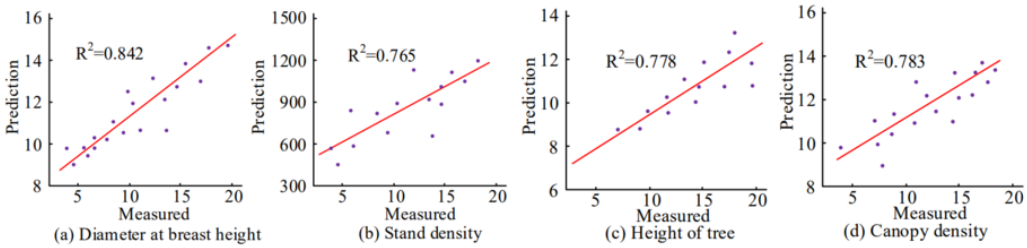
In the RF regression model, the parameters that required adjustment were number of trees (*n*tree) and variables per level (*mtry*), with *n*tree values ranging from 100 to 2000. The RF model was trained in fifty steps, and based on this, the *mtry* selection criteria were applied to optimize the *mtry* parameters for each forest stand factor, as illustrated in Figure 4. During the inversion of forest factors, the parameters associated with the model exhibiting the minimum RMSE were deemed optimal. According to the chart, the optimal *n*tree and *mtry* for canopy density were 150 and 6, respectively; for DBH, they were 300 and 8; for forest density, 100 and 3; and for tree height, 300 and 5.

Figure 4. The influence of parameters on the rf regression model



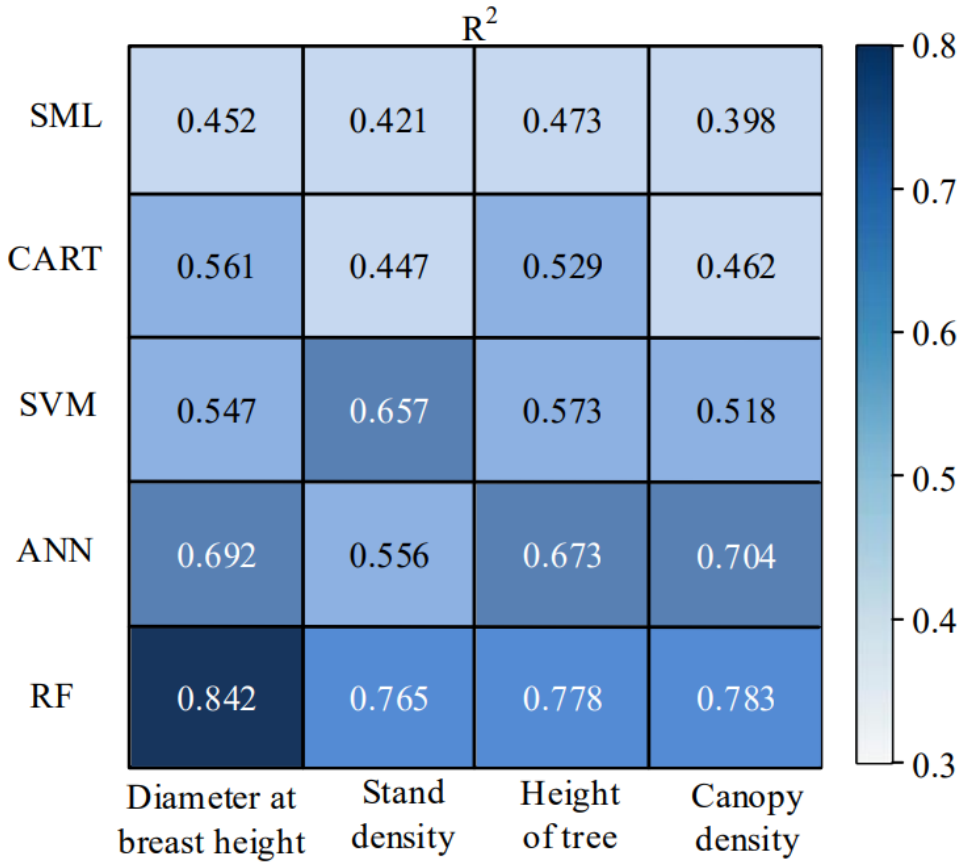
Using these optimal parameters, RF regression was performed on the stand factors, yielding an inversion accuracy for the stand factors that ranged between R^2 values of 0.66–0.85 and rRMSE values of 0.07–0.34. Among the various forest factors, the RF parameter inversion demonstrated the highest accuracy, with R^2 and rRMSE reaching 0.85 and 0.18, respectively. The inversion accuracy for several factors exceeded 0.80, with R^2 values of 0.83 and 0.81, and corresponding rRMSE values of 0.17, 0.18, and 0.15. Specifically, canopy density, crown width, and stand density all achieved inversion accuracies within the range of 0.77–0.78. These results indicated that the RF regression model provided high levels of accuracy, with forecast results that aligned well with the actual conditions.

Figure 5. Relationship between the RF regression model and stand factors



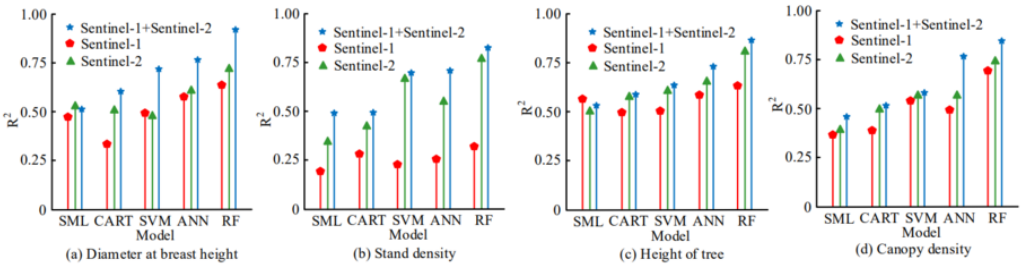
To identify the most effective model for forest factor inversion, the accuracy of various regression models, including stepwise multiple linear regression (SML), classification and regression trees, SVM and artificial neural networks (ANN), was compared. The comparative results are presented in Figure 5. As depicted, machine learning models generally outperformed linear models in terms of prediction accuracy, with SML showing the least performance among the machine learning models. Within the machine learning regression models, RF achieved the highest accuracy and the lowest error across all forest factors. ANN also exhibited high accuracy, second only to RF. The results highlighted that the application of RF in forest factor inversion yielded significantly higher accuracy than both ANN and SVM regression models. In contrast, classification and regression trees displayed lower accuracy and larger errors in the inversion of other parameters. Overall, the RF regression model emerged as a robust method for remote sensing inversion at a regional scale (Figure 6).

Figure 6. Precision comparison of different regression models



During the forest factor inversion process, the RF regression model was employed to integrate Sentinel-1 and Sentinel-2 data, resulting in the inversion of each forest factor. Generally, the joint inversion model using Sentinel-2 data provided higher precision when compared with the Sentinel-1 joint inversion model. In some cases, the inversion results from a single Sentinel dataset were more accurate than those from the combined data. The effectiveness of the combination was also evaluated. When comparing the performance of various regression models, however, the RF algorithm consistently demonstrated the highest precision in the inversion of various Sentinel data and forest factors (Figure 7).

Figure 7. Effects of different sentinel data on stand factors



DISCUSSION

The proposed model distinguished itself from similar works through several key aspects. It integrated multi-source data, combining ground sample data with Sentinel-1 SAR and Sentinel-2 multispectral imagery for a comprehensive forest quality assessment. Utilization of the Boruta algorithm for feature selection ensured that only significant variables were included, enhancing computational efficiency and predictive accuracy.

Incorporating texture information from Sentinel-1 SAR data via the gray level co-occurrence matrix captured structural characteristics of the forest canopy, providing additional insights into forest health not evident from spectral data alone. The optimization of RF regression models with specific parameters tuned for each forest stand factor achieved higher accuracy when compared with other models, such as SVM, ANN, and SML.

A detailed evaluation framework involving factor analysis refined the set of indicators used for forest health assessment, creating a weighted index system that highlighted the relative importance of various forest attributes. Focusing on the mountainous regions of western Beijing allowed for tailored modeling suited to local ecological conditions, further enhancing the relevance and precision of the assessments. Through these methods, the study provided an advanced approach to forest quality evaluation.

Integration of multi-source remote sensing data represents a promising avenue for enhancing the accuracy and comprehensiveness of forest quality assessments. While this study successfully utilized Sentinel-1 and Sentinel-2 data, future research could delve into the synergistic effects of combining these with other technologies such as UAVs and LiDAR. This would enable capturing both structural and spectral information of forests more effectively. Developing sophisticated models capable of handling diverse data types is essential for achieving this goal. Additionally, understanding how forest quality changes over time remains a significant challenge. Current analyses have been static; however, investigating temporal changes in forest attributes through time-series satellite imagery could offer insights into growth patterns, disturbance events, and recovery processes. Such an approach would be particularly beneficial for monitoring deforestation and reforestation efforts in regions like the Amazon.

The generalizability of the proposed method to other geographic areas poses another significant research direction. Although this study focuses on the mountainous regions of western Beijing, testing the approach in different ecological zones with varying forest types and climatic conditions is crucial. For instance, applying the methodology to tropical rainforests in the Amazon could provide valuable insights into deforestation dynamics and help guide conservation strategies. Moreover, further optimization of the RF regression model parameters, such as *n*tree and *mtry*, may yield improved prediction accuracies. Exploring alternative machine learning algorithms or hybrid models that combine strengths of multiple techniques could also lead to better performance metrics.

From a practical managerial perspective, the proposed approach offers substantial benefits for forest management by providing a cost-effective and scalable solution for assessing forest quality. Leveraging remotely sensed data enables forest managers to monitor forest health conditions in real-time, facilitating timely interventions against issues like pest infestations, disease outbreaks, and illegal logging activities. This is especially pertinent for large-scale forest management in regions such as the Amazon, where rapid deforestation poses a severe environmental threat. Accurate forest quality assessments are also critical for informing policy development aimed at sustainable forest management and carbon sequestration. High-resolution maps of forest attributes generated through this methodology facilitate targeted conservation efforts and assist policymakers in prioritizing areas for protection and restoration. Detailed information on forest degradation in the Amazon could support international agreements and initiatives aimed at reducing deforestation and promoting reforestation.

Beyond forest management, the methodology developed in this study holds broader application potential in fields such as agriculture, urban planning, and environmental monitoring. The integration of multi-source remote sensing data and advanced machine learning techniques can provide valuable insights into land cover dynamics, ecosystem services, and human-environment interactions. This adaptability enhances the utility of the approach for diverse stakeholders across various contexts and scales. By addressing these open research questions and highlighting the practical significance of the proposed methods, it becomes evident that there is considerable potential for advancing both scientific understanding and practical applications in forest quality assessment and management.

The proposed approach for assessing forest quality using a combination of ground sample data and satellite imagery faced several limitations. One significant limitation was the temporal discrepancy between the ground sample data collected in 2022 and the satellite imagery acquired in 2023. Despite some forest attributes remaining stable over short periods, significant disturbances such as wildfires or logging could have altered conditions, impacting the model's accuracy.

Spatial variability also posed a challenge. Forests exhibit considerable variation, which might not have been fully captured by the sampling strategy. Although stratified random sampling ensured representation across different strata based on an equal interval classification of forest land quality, localized variations or unique ecological niches within the study area might still have been missed.

Generalizability of models represented another limitation. Optimized RF regression models showed high performance metrics within the study area but may struggle when applied to other regions with different ecological conditions, tree species compositions, or management practices. Reliance on data from specific regions limits the applicability of findings to broader contexts.

Another limitation was that the study's focus on static analysis meant that the research did not capture dynamic processes of forest growth and site quality over time. This meant that the study did not account for seasonal changes, long-term trends, or the impact of environmental factors on forest health and productivity. Feature selection methods, while systematic, depended heavily on the initial dataset's quality and completeness. If important variables were missing or the dataset contained biases, critical indicators of forest quality might have been overlooked during the feature selection process.

Technological constraints further affected the assessment. Sentinel-1 and Sentinel-2 data, despite their power, had limitations related to resolution and spectral coverage. The inability of optical sensors to penetrate dense canopy cover potentially limited assessments of understory vegetation and soil properties. SAR data interpretation requires expertise and could have been influenced by atmospheric conditions and terrain effects. Addressing these limitations in future studies would require incorporating multi-temporal datasets, expanding geographic scope, integrating additional data sources, and developing more sophisticated models capable of capturing dynamic forest processes.

CONCLUSION

This study analyzed data from Sentinel-1 and Sentinel-2 satellites, which underwent preprocessing to remove thermal noise, correct orbital and terrain distortions, and perform image cropping. Remote sensing

parameters, including vegetation indices and slope, were extracted. A forest parameter and site quality inversion model was constructed to analyze the spatial relationship between forest quality and site quality. When using remote sensing data for the inversion of forest stand factors, the RF regression model achieved R^2 values ranging from 0.60 to 0.76, with RMSE values between 0.09 and 0.39. Optimal parameters for each variable were determined, leading to estimation accuracies for various forest parameters: average tree height and DBH had an accuracy of 75% to 80%, while canopy density and tree number density achieved an accuracy of 68% to 75%.

Forest land quality and site quality are dynamic processes, and this study primarily conducted a static analysis. Future research should explore multi-source remote sensing imagery further to investigate forest quality, site quality, and management efficiency. The integration of UAVs, LiDAR, and remote sensing imagery can enhance the accuracy of forest quality assessments, offering a more comprehensive and precise evaluation of forest attributes. Such advancements will improve overall assessment accuracy and contribute significantly to the field of forest management and conservation.

DATA AVAILABILITY

Specific details regarding data sources and access methods are provided below:

Sentinel Satellite Imagery

Sentinel-1

Provides C-band dual-polarization synthetic aperture radar images, which were downloaded from the European Space Agency's Copernicus Open Access Hub. Specifically, this study utilized ground range detected level SAR imagery acquired on June 25, 2023.

Sentinel-2

Offers multispectral imaging data, also retrieved from the Copernicus Open Access Hub. This research used Level-2A products obtained on August 10, 2023, with three different resolutions (10 meters, 20 meters, and 60 meters).

Auxiliary Datasets

To enhance the analysis, additional third-party datasets were integrated. These included digital elevation models sourced from the ASTER global digital elevation model (30-meter resolution), meteorological data from the China Meteorological Science Data Sharing Service Network, and soil type maps from the World Soil Information Service, provided by the International Soil Reference and Information Centre.

COMPETING INTERESTS STATEMENT

The authors of this publication declare there are no competing interests.

FUNDING STATEMENT

This research was supported by the Beijing Landscaping Youth Innovation Lift Project [grant number KJCX202311].

PROCESS DATES

Received: October 11, 2024, Revision: April 21, 2025, Accepted: May 13, 2025

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